ROLAND OOSTHUIZEN

Junior Frontend Developer (React) | Web Development & SEO

067 401 1290 • rolandoosthuizen09@gmail.com • NWU — BSc Computer Science & Economics (In Progress)

PROFESSIONAL SUMMARY

Junior developer with hands-on React experience on Orka Solar's in-house production system, plus three years of freelance experience designing, building, and optimizing WordPress and WooCommerce websites for real clients. Comfortable owning a project end-to-end — UI implementation, performance, SEO, and hosting — from first commit to live site. Outside of client and employer work, independently designed and built a full-scale Python machine-learning research system, reflecting a strong, self-directed interest in careful, well-tested engineering.

TECHNICAL SKILLS

- **Web Development:** React (JSX/TSX) • HTML5 • CSS3 • JavaScript • WordPress • WooCommerce
- **SEO & Performance:** Technical & on-page SEO • Google Search Console • Sitemap optimization • Site-speed optimization
- **Infrastructure:** Hosting management • Domain configuration • SSL setup
- **Data & ML (independent projects):** Python • pandas / NumPy • scikit-learn • SQLite • Streamlit • Plotly
- **Automation & APIs (independent projects):** LLM API integration (Groq) • Google Cloud TTS • MoviePy/FFmpeg • Pillow • REST API automation (Cloudinary, Instagram Graph API)
- **Other:** Basic 3D layout design (cable routing) • ERP data entry & management

PROFESSIONAL EXPERIENCE

Junior Developer — Orka Solar *July 2026 – Present*

- Work on an in-house company system built in React (JSX/TSX), contributing directly to a live production codebase.
- Resolve bugs and ship day-to-day UI improvements across the application.
- Design and build new feature templates as the system's functionality grows.
- Produce small 3D layout designs for cable routing to support system and site planning.
- Process and upload site load-profile data into the company's ERP system.

Freelance Web Developer & SEO Specialist — Self-employed *2023 – Present*

- Design and develop fully responsive websites using WordPress and custom CSS/HTML.
- Implement SEO best practices — meta structure, indexing setup, and sitemap configuration — and optimize load speed and mobile usability.
- Set up and manage hosting, domains, and SSL certificates.
- Build and customize WooCommerce stores, including product structure and filtering systems.
- Provide ongoing maintenance, debugging, and UI improvements for client sites.

KEY PROJECTS

Housey — housey.co.za

- Developed a modern, responsive website with mobile-first design principles.
- Implemented SEO structure and improved search visibility; optimized for faster load times across devices.

Drillpoint Africa — drillpointafrica.co.za

- Built and deployed a corporate website with full hosting configuration and SSL setup.
- Performed SEO optimization — sitemap submission, Search Console integration, and crawlability improvements.

WooCommerce & E-Commerce Builds

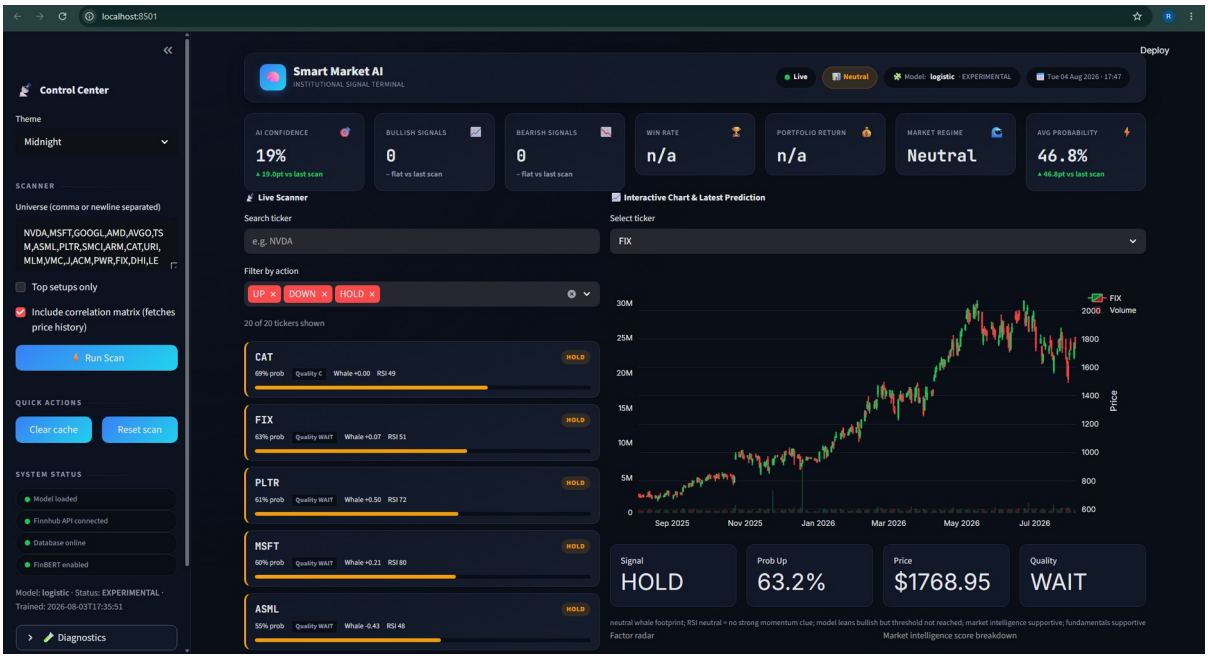
- Developed online stores with structured product categories and filtering systems, customizing layouts and checkout flow for usability and conversion.

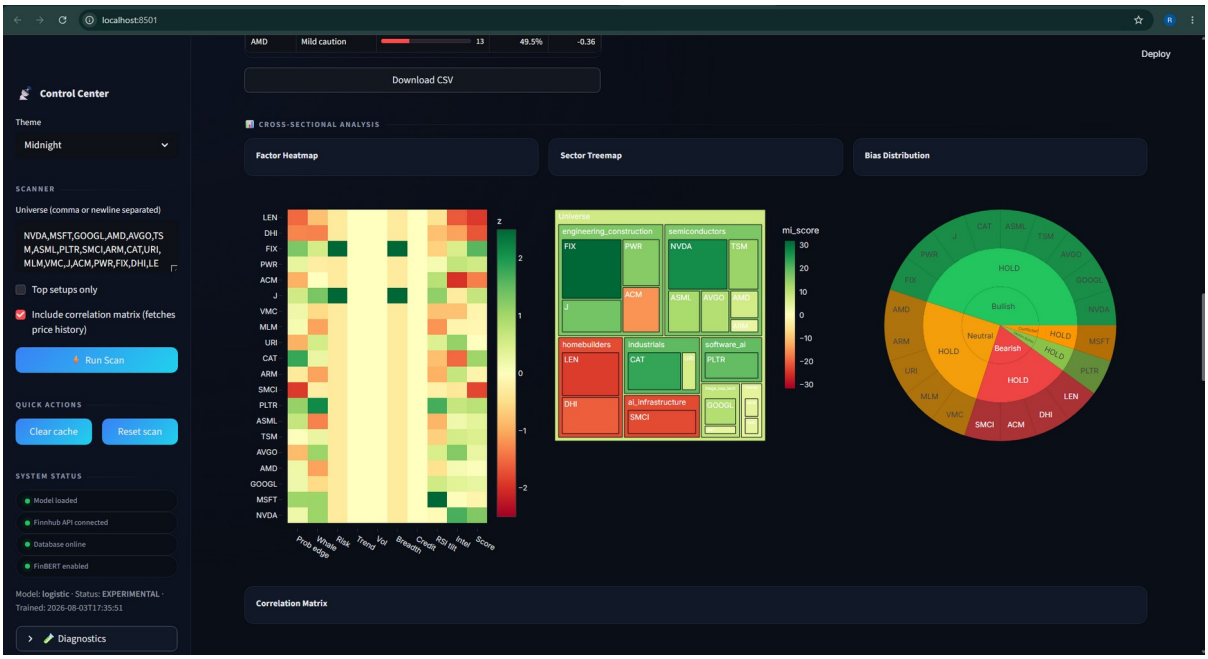
INDEPENDENT PROJECT — SMART MARKET AI (QUANTITATIVE TRADING RESEARCH SYSTEM)

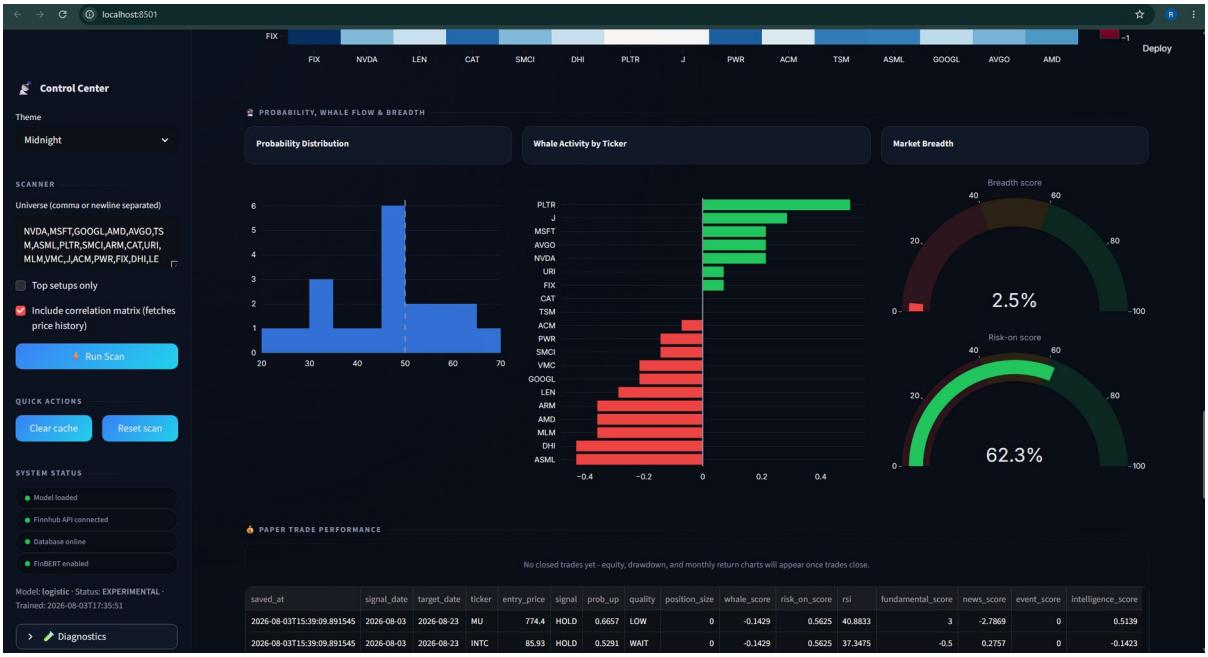
An independently built, end-to-end machine-learning system (Python) for researching systematic equity trading signals, developed and iterated on outside of coursework and client work. The project's focus was less on “does it make money” and more on building the discipline to honestly test whether it does — walk-forward validation, a held-out final

test period the model never sees during tuning, and a deployability gate that only certifies a model when it clears explicit accuracy, profit-factor, and drawdown thresholds on that untouched data.

- Built a walk-forward ML pipeline (scikit-learn: logistic regression, random forest, HistGradientBoosting, and an ensemble) with 58 engineered technical/market features, trained and compared across a rolling multi-year dataset.
- Implemented a regime-aware routing system that trains separate specialist models for bull, bear, high-volatility, mean-reversion, and neutral market conditions.
- Designed a five-part automated audit suite — feature redundancy, temporal stability, probability calibration, ticker/position concentration, and Monte Carlo trade-shuffle robustness testing — run on every training pass before a model is trusted.
- Added portfolio-level risk controls: ATR-scaled stop-loss/take-profit, daily trade and sector exposure caps, an equity drawdown breaker, and loss-cooldown logic.
- Built a closed-loop paper-trading and pattern-memory system that logs live predictions, checks real forward outcomes, and feeds results back for ongoing evaluation.
- Redesigned the research dashboard from a default Streamlit layout into a single-file, professional trading-terminal UI (Streamlit + Plotly) — candlestick charts, correlation matrices, sector treemaps, live signal cards, equity/drawdown curves, and model-confidence panels — while preserving every existing backend function untouched.







INDEPENDENT PROJECT — AUTOMATED SHORT-FORM VIDEO PIPELINE (PYTHON)

An end-to-end automation pipeline that ingests source content, rewrites it into narration scripts, and produces and publishes short-form vertical video without manual editing.

- Built a content-ingestion and filtering stage (RSS parsing, HTML cleanup, automated safety/content screening) feeding an LLM-based script rewriter (Groq API) with a custom system prompt and output-sanitization pass.
- Generated narration with Google Cloud Text-to-Speech, using SSML timing marks to sync on-screen captions precisely to speech, and composited vertical video programmatically (MoviePy/FFmpeg) with custom animated, emoji-aware caption rendering (Pillow).
- Automated the full delivery chain — video rendering, Cloudinary upload with retry/error handling, and publishing via the Instagram Graph API (media container creation, processing wait, publish).

EDUCATION

North-West University (NWU) — BSc Computer Science & Economics, In Progress (currently 1st year)

HS Ferdinand Postma — National Senior Certificate (Matric): Physics, Mathematics, Life Sciences

LEADERSHIP & EXTRACURRICULAR

Chess Team Captain — HS Ferdinand Postma (Grade 10 – Grade 12)

- Captained the school chess team for three consecutive years, combining strategic play with team leadership and organization.

Drama — HS Ferdinand Postma (Grade 9 – Grade 12)

- Participated in school drama for four consecutive years, Grade 9 through matric.